Remote Sensing Image Change Detection Transformer Network Based on Dual-Feature Mixed Attention

Xinyang Song, Zhen Hua, and Jinjiang Li

Abstract— **Change detection (CD) of high-resolution remote sensing (RS) images is a basic task in RS image processing tasks. In recent years, CD tasks have made many attempts in pure convolutional networks, attention mechanism, and transformer, and have achieved good results. Based on the power of attention and transformers, we hope to find a method that can handle the details of the image better and has better generalization ability. In this article, we propose a dual-feature mixed attention-based transformer network (DMATNet). First, we adopt a dual-feature extraction method, using a simple convolutional neural network (CNN) to extract coarse features, and a CNN based on progressive sampling to extract fine features. Then, we fuse the fine and coarse features with dual-feature mixed attention (DFMA) module. It can not only extract more specific regions of interest, but also overcome the misjudgment caused by oversampling, and synchronize feature extraction and target information integration. Finally, we use transformer to optimize these extracted information and feedback into the original features in the encoder to help remodel the pixel space. We merged the DMAT network into a deep feature difference-based CD framework and conducted extensive experiments on four datasets, LEVIR-CD, DSIFN-CD, WHU-CD, and CLCD, respectively, with tested F1 and interconnection over union (IoU) results of 90.75%/84.13%, 71.23%/55.32%, 85.70%/74.98%, and 66.56%/59.87%. Experimental results show that our DMAT-based model performs significantly better than the existing state-of-the-art attention and transformer-based methods.**

Index Terms— **Change detection (CD), convolutional neural network (CNN), dual-feature mixed attention (DFMA), high-resolution optical remote sensing (RS) images, Siamese network, transformer.**

I. Introduction

REMOTE sensing (RS) change detection (CD) is the process of comparing and identifying differences in the time dimension of RS images of the same region, which is crucial for understanding the detection of surface changes and detecting surface activity. The use of RS images to monitor and survey the surface has been widely used in a variety of applications, such as urban expansion and building CD, land use CD, disaster loss assessment, agricultural surveys, and map revision. Moreover, CD technology will greatly help us reduce the consumption of manpower and material resources. With the continuous development of Earth observation technology, it has attracted widespread attention [1] in recent years. Now the development of RS technology provides a large amount of high-resolution RS data, which provides a rich dataset for CD, but also puts forward new requirements for CD technology.

Optical RS images contain images of various factors such as various seasons, weather, and time. At the same time, with the development of modern times, the building complex has become more dense and complex. There are higher requirements for the task of image segmentation of high-resolution optical RS images. A good CD model can greatly reduce the influence of weather, lighting, and other factors on the processing results, so as to effectively identify the changed parts of the RS image. In general, a good CD model has the following capabilities: 1) efficient extraction and learning of rich feature information in high-resolution RS images and 2) to reduce the interference caused by pseudo-changes (parts that have not occurred or are not of interest).

Traditional CD methods can be divided into two categories: pixel-based CD and object-based CD according to the object being studied [2], pixel-based CD typically generates a difference map by directly comparing the spectral information or texture of a bitemporal image [3], [4], [5], [6], [7], [8], [9], The final result graph is then generated by methods such as threshold segmentation or clustering. But pixel-based analysis ignores information about the spatial context, so later techniques often detect changes based on objects. The main idea of the object-based approach is to extract features from the segmented image and identify changes that occur in the bitemporal image [10]. Celik et al. [4] apply principal component analysis (PCA) to image difference maps to extract representative features from pictures. In the post-classification method, the bitemporal image is independently classified and labeled. Some studies have begun to use the classification results for direct comparison [6], to extract the changed parts. To improve the performance of CD, the researchers viewed the CD problem as a problem of finding changes in land

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cover through a supervised approach. The most popular is the post-classification that adopts a supervised approach. However, these methods rely too much on empirically designed discriminant feature extraction algorithms and do not achieve satisfactory results on high-resolution images. Images presented in high-resolution images have increasingly high detail and more complex textures, creating new challenges for CD tasks.

In recent years, with the effect of deep learning in natural language processing (NLP), image analysis and other fields [11], [12], [13] have far exceeded the effect of traditional methods. RS CD tasks [14], [15], [16], [17], [18], [19] have also begun to widely use neural networks with fully convolutional neural network (FCN) [20] structures, and have achieved very good results. The depth-based feature-based approach converts a bitemporal image into spatial modeling information, using contextual information to detect changes in the part of interest. As [21] brings more convolutional layers into CD tasks, many networks also begin to add more convolutional layers to enhance the processing of features [21], [22], [23], [24]. Subsequently, Bahdanau et al. [25], the attention mechanism was applied to the field of NLP for the first time, and the attention mechanism began to become a hot topic for everyone to study. The attention mechanism has also brought inspiration to the field of RS, and the way in which the attention mechanism is applied to process RS images has also achieved remarkable results [21], [22], [26], [27], [28], [29], [30]. These attention-focused networks have better feature extraction methods than traditional convolutional-based processing methods, and are more effective at collecting global information.

There has always been a problem in CD tasks that cannot link global information in space-time, so researchers have begun to use attention mechanisms to enhance the processing of feature extraction [26], another way is to use attention to fuse information on a channel or spatial dimension with a bitemporal image and then weight features [27], [28], [29], [31]. At present, the pixel-based self-attention CD method [22], [23], [30] has achieved good results in detail by modeling the semantic relationship between pixels in space, but this calculation efficiency is very low and the computational complexity will continue to increase. With the success of transformers [32] in NLP, a large number of research began to apply transformers to various computer vision tasks, and many architectures were born for different visual tasks. For example, vision transformer (ViT) [33], Segmentation Transformer (SETR) [34], ViT using Shifted Windows (Swin) [35], Twins [36], and SegFormer [37]. These transformer networks achieve better results than deep convolutional neural network (CNN) in many fields, with better feature extraction and the ability to model contexts. With the remarkable results achieved by transformer in various work, the processing of hyperspectral RS images has also begun to show interest in transformers, and CD tasks have begun to try to use transformers to process RS image features [38], [39], and good results have been achieved, but there is still a lot of room for improvement in the processing of features and the depiction of details.

In this article, we propose a transformer RS image CD network based on dual-feature mixed attention based transformer network (DMATNet) for CD of high-resolution bitemporal RS images, as shown in Fig. 1. In our network framework, first, we process the bitemporal image with two different roles of CNN networks, where the backbone part is processed by the CNNA progressive sampling module (PSM). The PSM can continuously collect and update picture information (including location information and feature information) step by step through iteration, and mark the parts of interest to reduce the interference of the parts that are not of interest. These continuously sampled information will communicate with the feature information extracted from the CNNB, and then further optimize the sampled features, leave useful information behind, and weaken the weight of useless information. Second, these remapped information is extracted through the sample tokenizer module into several compact high-level semantic tokens that contain the main feature information extracted from the CNN backbone. Finally, after this information is optimized by the transformer encoder (TE), the original feature extracted from CNNA is connected in the decoder, the context information is used for modeling, the feature picture of the bitemporal image is output, and the feature difference map of the bitemporal image is calculated in the prediction head.

Our work contributions can be summarized as follows.

1) During the sampling process, a PSM is utilized to optimize the transformer-based RS image CD framework. The PSM module utilizes an iterative approach for progressive sampling (PS), by which the sampling area is continuously refined so that the final region of interest is focused on the target region, thus allowing the results to generate prediction maps with more detailed edges. This has greatly improved our ability to handle details.
2) To balance the neglect caused by too deep sampling and the blurred edges caused by too shallow sampling, we propose for the first time a dual-feature mixed attention (DFMA) module using the interaction of two feature information, which can fuse the sampled fine features with the coarse features to extract more accurate and fine information. This largely compensates for the possible omission due to over-sampling and unclear edges due to a shallow CNN alone in PSM.

The rest of this article is organized as follows. Section II describes the work related to the development of the CD framework. Section III describes in detail the method we have proposed. To verify the effectiveness of our method, we did some experiments in Section IV and discussed the method we proposed. In Section V, we discuss the strengths and weaknesses of our approach and the directions for future improvements. Finally, in Section VI we summarize all the work in this article.

II. Related Work

A. CD Method Based on Deep Learning

Compared with the original traditional CD method, deep learning methods have achieved rich experimental results in recent years. Most of these methods consist of two steps: unit analysis and change identification [1]. Analysis unit, as a major step in the change monitoring framework, aims to

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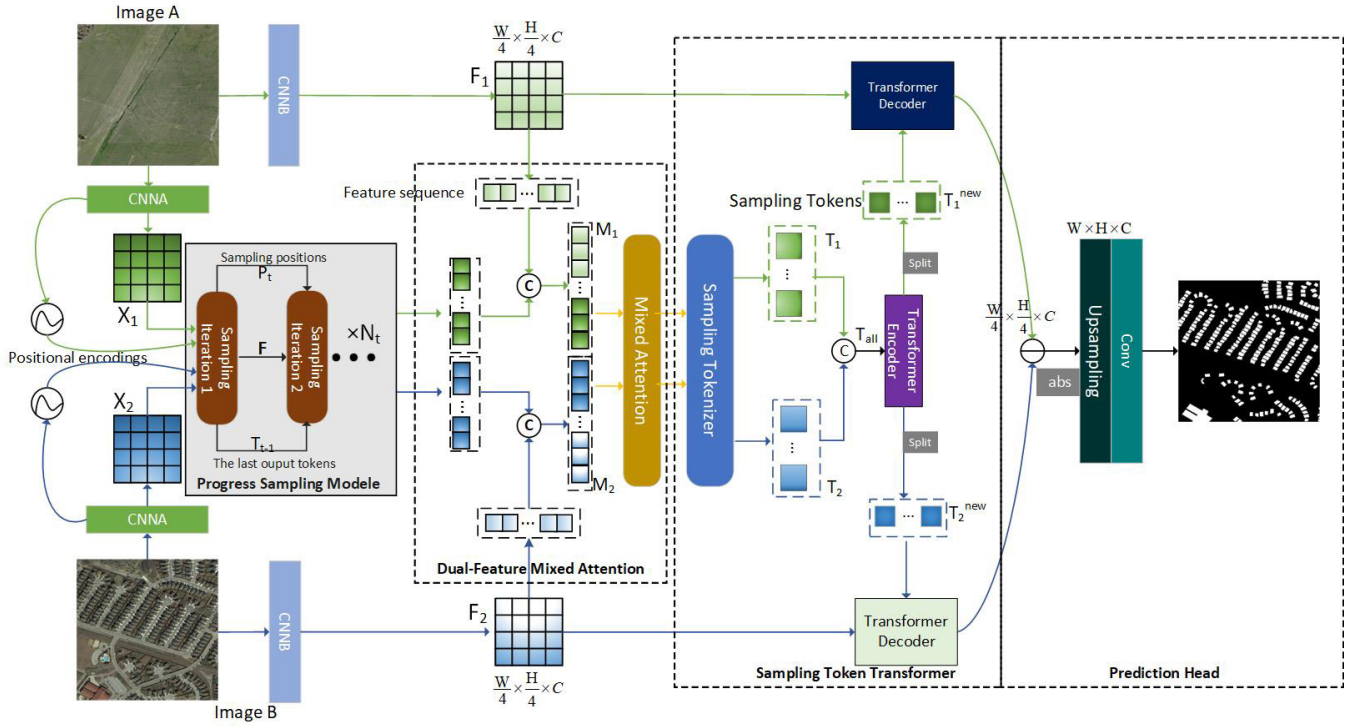


Fig. 1. Description of our DMATNet model. These include the dual-CNN backbone, PSM, DFMA module, and sample tokenizer transformer module. The PSM performs sampling operations in an iterative manner, the DFMA module can fuse the feature information of different feature extraction methods, and the ST transformer module extracts several compact semantic tokens from the feature information, and uses the transformer to process and finally map into the original feature. The final prediction image is produced at the prediction head.

establish a link between the raw data information from the cell. Deep learning CD methods based on unit analysis are divided into three types: 1) pixel-based methods; 2) object-based methods; and 3) depth-based feature-based methods.

The pixel-based method first converts the picture into depth eigenvalues through a neural network, and these eigenvalues are detected by depth feature selection and pixel-level comparison. For example, Zhang et al. [40] use a deep neural network to separately extract feature information from a bitemporal image, and then map the feature map of the bitemporal change to a 2-D polarity domain to represent the changed information. Saha et al. [41] also used a deep CNN network for feature extraction, and then compared the feature map of the change by analyzing the bad vector. Hou et al. [42] and Peng and Guan [43] used the Visual Geometry Group (VGG)-16 framework for feature extraction of bitemporal images.

The object-based method first uses segmentation technology to extract the segmented image, and then uses the segmented object as an analysis unit for change monitoring. Several steps are usually followed: image segmentation, feature extraction, feature difference comparison, and clustering cargo threshold classification. For example, Lei et al. [44] propose a stacked denoising autoencoder on the result of a split. Amin et al. [45] used a pretrained local CNN framework to separately extract and link features from multiple regions for deep feature extension to solve the finite features of single superpixel. Although the above two methods utilize deep learning techniques, they can effectively transform the original image into a high-level feature space and retain key information. However, error propagation still exists, and there are disadvantages such as empirical threshold determination as traditional CD methods. The CD task began to introduce a FCN, a CD method based on depth features. Long et al. [20] first proposed an FCN for image segmentation tasks, enabling end-to-end classification. Most of the existing pixel-level depth feature CD methods [15], [21], [22], [23], [23], [26], [27], [28], [29], [30], [46], [47], [48], [49], [50], [51] use FCNs to directly use two input images to generate high-resolution change maps. Existing FCN-based approaches can be roughly divided into two groups based on the stages of bitemporal information fusion. One is to pass the bitemporal image into a semantic segmentation network as a single input [23], [46], [47], [48], and the other is to use the neural network to first extract the features of the bitemporal image, and then predict the difference image based on the fusion features [15], [21], [22], [23], [23], [26], [27], [28], [29], [49], [50], [51].

Most of the current work, to improve the identification ability of neural networks as the main task, through the design of multilevel feature fusion structure, combined with segmentation technology, as far as possible to increase the receptive field (RF) so that the model has better spatial and temporal context modeling capabilities. With the increasing complexity of satellite images and the increasing accuracy requirements, in addition to FCN networks, existing technologies also utilize attention mechanisms, such as channel attention [26], [27], [28], [29], space attention [26], [27], [28], self-attention [22],

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[23], [30]. However, these efforts to establish a contextual connection between space and time either by placing the image into a feature enhancement module [26] or by directly weighting the features in the channel or spatial dimension [27], [28], [29]. Recently, the proposed super-resolution-based CD network [52] proposes a new direction for CD, which utilizes the way of stacking attention modules to process information.

In addition to this, many recent studies have taken an interest in deep supervision [53] and adversarial learning [54], [55]. For example, Li and Huo [55] used generative adversarial training to generate more variable bitemporal images involving a large number of different buildings. Zhang et al. [54] use cycle-consistent adversarial networks to learn subimage-to-subimage mapping, using a given heterogeneous image pair for CD. However, this method requires a lot of training, and the training efficiency is relatively low.

In hyperspectral images, to better identify target regions, many studies [56], [57] have sampled methods such as multistructural unification or multitasking to enhance the recognition for hyperspectral images. In addition, [58] proposes a pyramid-based fully convolutional network. The multiscale variation information is further extracted from the difference feature maps processed by the encoder structure of the original network. There is also a part of research [27], [59] want to improve the effectiveness of the method using supervised learning. All of these methods have made effective attempts at CD.

To better improve the detail portrayal for the CD dataset, this article proposes a high-performance attention CD network that maintains the original features and further fuses high-level feature information. Unlike existing methods based on deep learning and attention mechanisms, the former is to directly model the dense relationship based on any different feature information elements in pixel space, while we use the attention mechanism to fully exchange information with the features extracted by the two extraction methods, and then aggregate the features with rich information containing dual-feature processing into several compact semantic token blocks, using these tokens to guide the modeling of the original feature information. Our intuition is to utilize a DFMA module to extract features with more spatiotemporal information, so that the semantic tokens extracted later contain more contextual information, which can summarize the information about changes in interest in the scene, greatly improving the modeling ability of the original features of pixel space.

B. Transformer-Based Model

With the introduction of transformer [32] in 2017 and used for machine translation tasks, it surpassed previous sequence models based on complex recursion or CNNs. Subsequently, in the field of NLP, various models based on transformers have also achieved good results, which has led to a trend in the field of computer vision to study transformers, such as image generation [60], [61], image classification [33], [62], [63], object detection [24], [64], [65], image subtitles [66], segmentation [34] and super resolution [67], [68], and many other visual fields. Standard transformer blocks consist of a multihead self-attention (MSA), a multilayer perceptron (MLP), and a layer normalization (LN). MSA play a key role in establishing global dependencies between input and output sequences. Transformer models have achieved amazing results in the field of vision, which has to make the field of RS also explore the advanced functions of transformer models. At present, there have been many attempts in the field of RS, for example, image time-series classification [69], [70], RS image captioning [71], [72], scene classification [73], and hyperspectral image classification [74]. He et al. [74] applied a variant of the transformer (bidirectional encoder representations from transformers (BERT) [75]) to capture global dependencies between pixels in hyperspectral image classification. In addition, the CNN transformer method proposed by Li et al. [70] can use the transformer to learn semantic information from the multitime feature sequence extracted from the CNN, which can classify the time series image.

While achieving good results in many areas of RS images, CD has also begun to experiment with transformer-based models. Chen et al. [38] using the transformer's ability to learn bitemporal images, extracted some semantic markers from the graph and modeled the context with the help of markers in space time. Feng et al. [39] proposed a cross-interaction and interscale feature fusion network within the scale, which clearly excavated the fusion capabilities of CNN and transformer. This gives the CD task a different ability than the previous one based solely on deep CNN and attention mechanisms.

Inspired by PS-ViT [76], we added a PS model with better sampling capabilities to the transformer model, which can solve the problem of insufficient and complex sampling using convolutional networks alone. Moreover, we exchange information with the simply extracted advanced features after sampling, avoiding erroneous information extraction while extracting valid features. In this article, we explore the performance of the PS module and the DFMA module into the CD network. Part IV proves that the network model we propose is more efficient in global semantic relations than the recently advanced CD network, and has better feature extraction effect than the current transformer-based network.

III. METHODOLOGY

In this section, we will go into detail about our DMATNet model, as shown in Fig. 1. We introduce a DFMA and transformer into a CD framework because we wanted to take advantage of the mixed attention mechanism and the feature extraction that transformer brings.

Several major components of our network include: 1) PSM, which employs an iterative and PS strategy to locate the discriminant region, and in the iteration uses the results of the previous one to jointly update the next sampling result; 2) DFMA, this synchronous modeling scheme enables the two feature extraction methods to communicate the main features of a specific target extensively, increase the weight of the effective features, and avoid the interference caused by not being interested; and 3) sampling tokenizer (ST), used to

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extract compact semantic tokens containing rich contextual information, to help model context in space-time.

The algorithmic inference process of our overall framework, as shown in Algorithm 1.

Algorithm 1 Implementation Process of Our DMATNet Model

Input: A, B $(bi-temporal\ image)$)
Output: $P(a\ prediction\ change\ mask)$
// $step1: Extraction\ of\ dual\ features$
$\mathbf{X_1} = CNNA(\mathbf{A}), \mathbf{X_2} = CNNA(\mathbf{B})$
$\mathbf{X_1} = CNNB(\mathbf{A}), \mathbf{F_2} = CNNB(\mathbf{B})$
for $i\ in\ \{1,2\}$ **do**
$\mathbf{F_i} = PSM(\mathbf{F_i})$
end
// $step2: Establish\ a\ link\ between\ the\ two-part\ feature$
// $Extract\ the\ exchanged\ inormations$
$\mathbf{M_1} = Concat(\mathbf{F_1}, \mathbf{X_1}), \mathbf{M_2} = Concat(\mathbf{F_2}, \mathbf{X_2})$
for $i\ in\ \{1,2\}$ **do**
$\mathbf{M_i^{new}} = DFMA(\mathbf{M_i})$
$\mathbf{F_i^{new}} = split(\mathbf{M_i^{new}})$
$\mathbf{T_i} = ST(\mathbf{F_i^{new}})$
end
// $step3:\ use\ encoder\ to\ generate\ context-rich\ tokens$
// $use\ decoder\ to\ refine\ the\ original\ features\ X$
// $Encoder$
$\mathbf{T_{new}} = Transformer\ (\mathbf{T})$
$\mathbf{T_1^{new}}, \mathbf{T_2^{new}} = Split(\mathbf{T_{new}})$
//$Decoder$
for $i\ in\ \{1,2\}$ **do**
$\mathbf{X_i^{new}} = Transformer\ Decoder(\mathbf{X_i}, \mathbf{T_i^{new}})$
end
// $step4: Calculate\ the\ feature\ difference\ plot\ and\ generate\ masks\ from\ the\ prediction\ head$
$\mathbf{P} = Prediction\ Head(|\mathbf{X_1^{new}}, \mathbf{X_2^{new}}|)$

A. PS Module

To pay more attention to the parts of the image that are of interest, and to mitigate the problem of structural damage, we introduce a PS module to sample the image to preserve more efficient features, as shown in Fig. 2. The PS module is an iterative framework. For a feature map $F \in \mathbb{R}^{C\times H\times W}$ (C, H, and W represent the size, height, and width of the feature channel, respectively), we divide the input picture rules into patches of 64 × 64, and these patches will not consider their own importance to the overall content, but only consider the part of interest within the patch to reduce the damage to the structure. The features that have been sampled stepwise are expanded into a token $T \in \mathbb{R}^{C\times(n\times n)}$, $(n \times n)$ representing the number of samples in a graph, and N is the total number of iterations in the PSM. Each iteration of PS will extract the current sampling position, and use the position information of the previous iteration to update the sampling result, the iterative process can be formally expressed as

$$p_{i+1} = p_i + o_i, \quad i \in \{1, \ldots, N-1\} \tag{1}$$

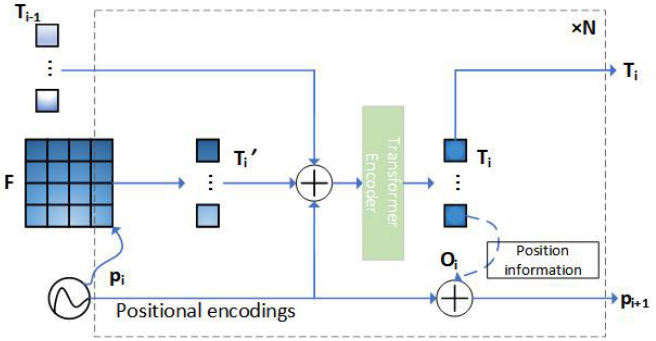


Fig. 2. Depicts the PSM. F represents feature information, p_i represents the position information at iteration i, T_i represents the information after sampling at iteration i, and T_{i-1} is the sampling information at $i-1$ iteration, o_i representing the offset matrix predicted at iteration i.

where $p_i \in \mathbb{R}^{2\times(n\times n)}$ represents the sampling position matrix at iteration i, and $o_t \in \mathbb{R}^{2\times(n\times n)}$ represents the offset matrix predicted at iteration i. At each iteration, the position information is passed to the picture feature, and according to the mapping of the position code on the input feature, the sampling information after each iteration is sampled, and the sampling process can be expressed as

$$T_i' = F(p_i), \quad i \in \{1, \ldots, N\} \tag{2}$$

where $T_i' \in \mathbb{R}^{C\times(n\times n)}$ is the initial sampling marker at iteration i. F stands for the input feature image, the sampling position at p_i. To completely map the p_i in the form of fractions into the feature, we use a bilinear interpolation operation to achieve this step, and this operation is differentiable for both the input picture feature F and the sampling position p_i. The initial sample marker, the output marker of the last iteration, and the position encoding of the current sampling position are concated, and the output tokens produced by the current iteration are obtained after passing through a TE. Formally, we denote it as

$$\begin{aligned} P_i &= W_i p_i \\ F_C &= F_i' \oplus P_i \oplus F_{i-1} \\ F_i &= \text{TE}\ (F_C), \quad i \in \{1, \ldots, N\} \end{aligned} \tag{3}$$

where $W_i \in \mathbb{R}^{2\times C}$ is a linear transformation of the sample position p_i projected to the position encoding matrix P_i of size $C \times (n \times n)$, and W_i assists in each sample location mapping. TE stands for multihead self-attention-based TE layer. In the process of sampling, our sampling positions are unequal, and we use the normalized absolute coordinates of each sample point. These coordinates will correspond to different locations in the feature, embedded in the corresponding location to help extract the feature, and help predict the sample location offset for the next iteration, in the form

$$o_i = M_i T_i \quad i \in \{1, \ldots, N-1\} \tag{4}$$

where $M_i \in \mathbb{R}^{2\times C}$ is a learnable transformation of the prediction offset matrix.

The sampling method of PS can help the feature convergence position to gradually shift to the part of interest, which can reduce the error when sampling as much as possible.

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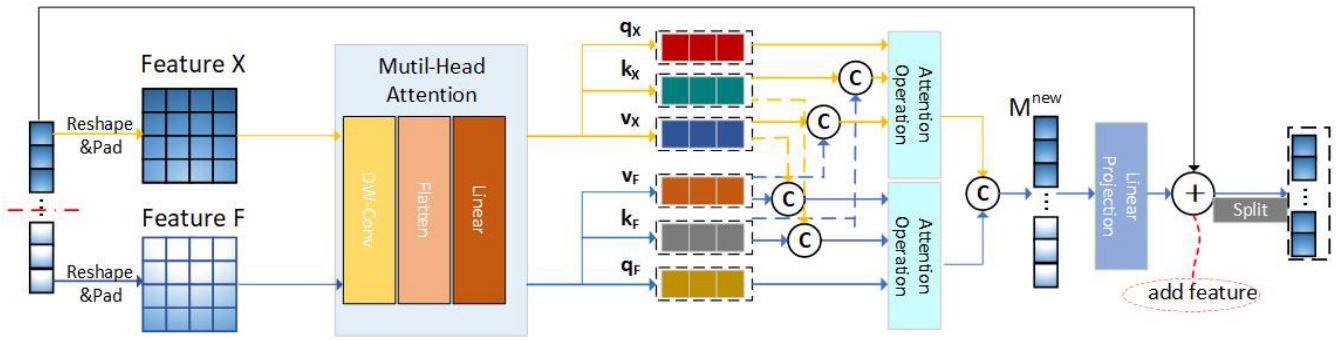


Fig. 3. Depicting DFMA. The module includes self-attention and cross-attention, in which both feature extraction and information integration are performed for the results of the two CNN methods, which can avoid the negative effects of potential interferences as much as possible.

B. Dual-Feature Mixed Attention (DFMA)

DFMA is an important part of our proposed CD network to extract more precise segmentation targets. The input to DFMA is image feature F and image feature X extracted by CNN, which aims to extract their common features at the same time and fuse the interaction information between them. Unlike attention in other CD tasks, DFMA can perform double-attention operations on two sets of independent token sequences in two feature sections. First it focuses on the token itself in each sequence to capture information from the two sequences, while it cross-notes the tokens of the two sequences, which is the exchange of information between the two sequences. This is shown in the Fig. 3.

First, given a merge tokens containing two different sequence tokens, after entering the DFMA module, we split it into two parts and remodeled them into 2-D feature mappings. To help model the local spatial context of the two sequence tokens, for each feature map we used a separable depth-wise convolutional projection layer (i.e., query, key, and value). To improve efficiency, we also downsampled the key and value matrices. The two feature plots are then flattened and processed by a linear projection to produce queries, keys, and values for the attention operation. We use q_X, k_X, v_X and q_F, k_F, v_F, respectively, to represent the query, key, and value of feature X and feature F. For the generated tokens, we mix them with interest, which can be defined as

$$\begin{aligned} k_{XF} &= \text{Concat}(k_X, k_F), \quad v_{XF} = \text{Concat}(v_X, v_F) \\ \text{Attention}_X &= \text{Softmax}\left(\frac{q_X k_{XF}^T}{\sqrt{d}}\right) v_{XF} \\ \text{Attention}_F &= \text{Softmax}\left(\frac{q_F k_{XF}^T}{\sqrt{d}}\right) v_{XF} \end{aligned} \tag{5}$$

where d represents the dimension of the key, Attention_X and Attention_F representing the attention map generated for feature X and feature F, respectively. The attention mechanism here includes self-attention and cross-attention, in which both feature extraction and information integration are carried out for the results of the two CNN methods, which can avoid the negative effects of potential interferences as much as possible. At the same time, since our high-resolution RS dataset contains a large number of fine features, to reduce the computational cost of DFMA, we hope to use an effective module to process the extracted information to simplify the amount of information required for operation and retain only important information. Therefore, we take an asymmetrical mixed-attention module, defined as follows:

$$\begin{aligned} \text{Attention}_X &= \text{Softmax}\left(\frac{q_X k_{XF}^T}{\sqrt{d}}\right) v_{XF} \\ \text{Attention} &= \text{Softmax}\left(\frac{q_F k_F^T}{\sqrt{d}}\right) v_F \end{aligned} \tag{6}$$

we use a mix of key and value for feature X to focus on it, while feature F is only self-focused. It is worth noting that feature X, which has been extracted by PS, is a key part of passing information backwards, so we need to focus on the global impact of feature X. For feature F, it is just to help make the extracted information more efficient. In this way, we try to keep the extracted feature map as much as possible on the basis of maintaining the original features, reducing the impact on other nonessential feature parts.

Note that our DFMA module has a similar mechanism of attention, however, we convolve the spatial structure information into a deep-wise convolution when using position encoding. And the feature extraction and information integration included in our DFMA module are multistage backbones and differ from the attention mechanism of a separate CNN backbone for extraction. In the end, we use a mixed-attention mechanism, which can further improve our efficiency in extracting valid information while ensuring precision.

C. Sampling Tokenizer

For high-resolution RS images, if all the sampled feature information is transmitted in the network, such a consumption is huge. Therefore, we hope that feature information can be summarized with some high-level conceptual information, which includes not only the characteristics of interest in the picture, but also spatial information. Here we introduce a ST module, as shown in the Fig. 4. It is used to extract compact semantic tokens that represent features of interest and corresponding spatial locations, and the semantic concept information in the token can be mapped to the original features to help model the original features, and these semantic information can be shared by bitemporal images. Our intuition

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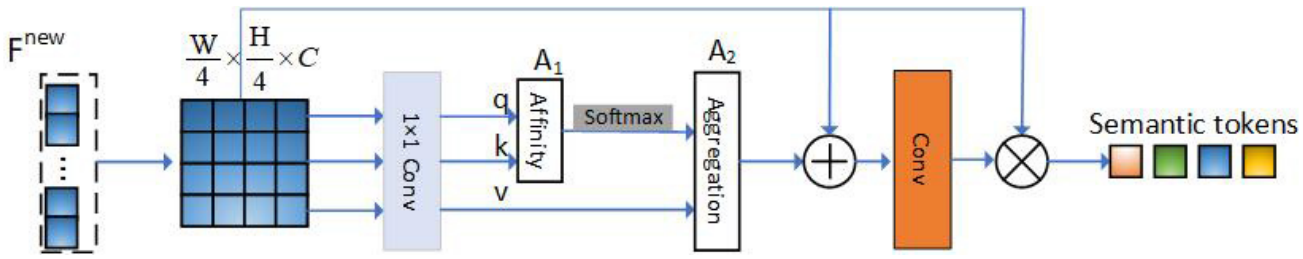


Fig. 4. Depicts the Sample Tokenizer. Extract features into compact semantic tokens that represent features of interest and corresponding spatial locations.

is that for the features of the input $F^{\text{new}} \in \mathbb{R}^{C\times(n\times n)}$, after going through the ST module, a large amount of picture information can be summarized into several visual words, each word corresponding to a marker vector, this method is similar to the tokenizer in NLP.

We first process the imported image features ${F_1}^{\text{new}}$, ${F_2}^{\text{new}} \in \mathbb{R}^{C\times(n\times n)}$ in equal proportions into the form of $I_{\{1,2\}} \in \mathbb{R}^{C\times H\times W}$. Since convolution operations only operate on pixels in the sensory field, it has certain limitations, and it is difficult for common FCN networks to extract contextual semantic information and global features. Therefore, we employ a criss-cross attention (CCA) module to improve the semantic comprehension of the model. For the input features I, the dimension reduction is obtained by two convolutions of 1×1 to obtain q, k. For each location u of the spatial dimension in q, we can get a vector $Q_u \in \mathbb{R}^{C'}$, C' is the number of channel, which are in the same row or column with position u. At the same time we can obtain a set $\Omega_u \in \mathbb{R}^{(H+W-1)\times C'}$ by extracting from k and the feature vectors with the position u in the same row and column, and perform affinity operations on them

$$d_{i,u} = Q_u \Omega_{i,u}^T \tag{7}$$

where $d_{i,u} \in \mathbb{R}^{(H+W-1)\times(W\times H)}$ is the degree of correlation between features Q_u *and* $\Omega_{i,u}$, $\Omega_{i,u}$ represents the ith element of Ω_u, $i \in [1, \ldots, H+W-1]$. Then the attention graph A_1 is calculated by a softmax of a channel dimension. I generates v through another 1*1 convolutional layer, we can get a vector v_u and the set Φ_u, Φ_u is the feature vector in v that is along with or in the same column as the position u, and then the context information is collected through an aggregation operation, and finally the final result is generated with the input features I. Formally

$$I_u^{\text{new}} = \sum_{i=0}^{H+W-1} A_{i,u}\Phi_{i,u} + I_u \tag{8}$$

where I_u^{new} is the feature vector in output at position u, and $A_{i,u}$ is the scalar value in channel i and position u in A_1.

New features I_1^{new}, $I_2^{\text{new}} \in \mathbb{R}^{C\times H\times W}$ can be obtained through CCA. The CCA module aggregates the length and width from the context separately, so that each pixel can be contacted so that there is pixel information. For each pixel I_i^p on feature image I_i^{new}, we put it into a point convolution module to get L semantic groups, and then use softmax to calculate the spatial attention mapping of the H and W dimensions. Finally, we map the above calculated result into the initial feature I_i, and obtain two sets of tokens $T^1, T^2 \in \mathbb{R}^{C\times L}$, by calculating the weighted average of pixels I_i^p, where L is the size of the vocabulary set of tokens. Formally

$$\begin{aligned} I_i^{\text{new}} &= CCA(I_i) \\ T^i &= (A)^T I_i = \left(\sigma\left(\emptyset\left(I_i^{\text{new}}; W\right)\right)^T I_i, \quad i \in \{1, 2\} \end{aligned} \tag{9}$$

where $\emptyset(\cdot)$ denotes the point-wise convolution with a learn-able kernel $W \in \mathbb{R}^{C\times L}$, $\sigma(\cdot)$represents a softmax operation that obtains attention map $A \in \mathbb{R}^{HW\times L}$. T_i is calculated by calculating A and picture feature I_i is calculated by multiplication.

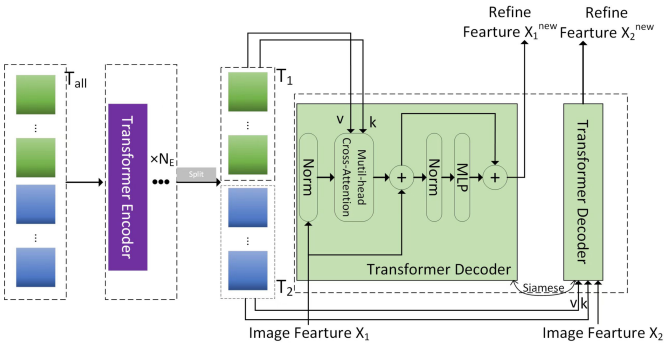


Fig. 5. Describes our TE and TD.

D. CD Model Based on Transformer

We incorporated Transformer into a CD framework, as shown in the Fig. 5. In transformer, we tie the extracted tokens and encode information about the corresponding or absolute position of the element in marker-based space-time with the help of an encoder. The encoded information will be projected in the decoder into pixel space, so that each pixel gets the required weight to achieve the purpose of refining the feature mapping.

1) Transformer Encoder: To better obtain contextual information, we use a TE for the two semantic tokens T_1, T_2 we obtain. We can use the TE to take advantage of the global semantic relationship of these tokens in space-time, encoding these tokens based on contextual spatiotemporal information. Here we connect the tokens generated by the two images together with a concat operation to form a token collection $T \in \mathbb{R}^{2L\times C}$. T processes the input encoder to obtain a new set of tokens $T^{\text{new}} \in \mathbb{R}^{2L\times C}$. Finally, we divide the new token set into two separate tokens T_1^{new},T_2^{new} in the same format as the original. TE mainly consists of a MSA block module and a MLP module. Unlike the original transformer that used

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the post-norm residual unit (PreNorm), we use the prenorm residual unit, i.e., we normalize the layers before MSA/MLP. At each layer, the set of input tokens, T, splits the query, key, and value, and then passes three values into the self-attention. Formally expressed as

$$Q = T \cdot W^q, \quad K = T \cdot W^k, \quad V = T \cdot W^v$$
$$h = A(Q, K, V) = \sigma\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{10}$$

where W^q, W^k, W^v are the learnable parameters of the three linear projection layers, C represents the number of channels, and d represents the dimensions of the three linear projections. h is the output of the attention head in the MSA, i.e., the result of adding attention to the three linear projection layers Q, K, V. A represents self-attention and $\sigma(\cdot)$ represents the softmax function that operates on the channel dimension.

At the core of the transformer is MSA. The MSA executes multiple independent attention heads in parallel and connects the outputs to project back to the final value. The operation of the MSA can be expressed as

$$\text{MSA}(T) = \text{Cat}(h_1, h_2, \ldots, h_n)W^m$$
$$\text{MLP}(T) = \text{GELU}(TW_1)W_2 \tag{11}$$

where $h_i (i = 1, 2, \ldots, n)$ represents the output of each attention head, and $W^m \in \mathbb{R}^{nd\times C}$, $W_1 \in \mathbb{R}^{C\times 2C}$, $W_2 \in \mathbb{R}^{2C\times C}$ is the linear projection matrix.

2) Transformer Decoder: Using the TE module, we extract new tokens rich in contextual information for each time image. These tokens can better reveal the interest transformation. To obtain pixel-level features, we need to project these conceptual information back into pixel space to refine each temporal feature. We use an improved transformer decoder (TD) to achieve this. After serialization of the extracted image feature X_i, a new sequence feature X_i^{new} is obtained, which utilizes token T_I^{new} in the transformer to generate a new fine feature. We use the pixel in X' as the query and the resulting token as the keys. In the decoder, we avoided the massive computation caused by too dense information between pixels, we removed the MSA block, retaining the multihead cross attention (MA) in the encoder and the MLP module with the same configuration as the encoder. Queries in MA derive from image sequence features X_i^{new}, keys and values come from tokens T_i^{new}. Formally, a MA module can be represented as

$$\text{MA}\left(X_i', T_i^{\text{new}}\right) = \text{CAT}(h_1, h_2, \ldots, h_n)W^m \tag{12}$$

where $h_i (i = 1, 2, \ldots, n)$ represents the output of each attention head, and $W^m \in \mathbb{R}^{nd\times C}$ is linear projection matrix.

E. Other Detail of the Network

1) CNN Backvone for Dual-Feature Extraction: For bitemporale RS images, we process each image using two improved ResNets, one is the improved ResNet18 to generate image feature X, and the other is the improved ResNet50 to generate image feature F. The original ResNet had five stages, and each stage had two downsamplings. Our ResNet18 reduces the amount of information by setting the stride of the next two stages to 1 and adding a convolution operation with a channel count of 32 and a bilinear interpolation operation with a sampling factor of 0.25 at the end. For the improved ResNet50, we only used the first phase of the convolutional stem and two bottleneck as feature extraction modules.

2) Prediction Head: After our PSM and DFMA module, we extracted high-level semantic features, so we only need to perform a simple FCN at the end for change recognition. First, we upsample the output from the transformer to generate two feature maps $X_1^*, X_2^* \in \mathbb{R}^{H_O\times W_O\times C}$ (H_O, W_O represents the height and width of the original image, respectively), and then generate a predicted change probability plot $M \in \mathbb{R}^{H_O\times W_O\times 2}$ in the prediction head, which can be represented as

$$M = \sigma(g(D)) = \sigma\left(\text{g}\left(|X_1^* - X_2^*|\right)\right) \tag{13}$$

where $D \in \mathbb{R}^{H_O\times W_O\times C}$ represents a feature difference image (FDI) and is achieved by two feature plots X_1^*, X_2^* by element-level absolute subtraction, $g : R^{H_O\times W_O\times C} \Rightarrow R^{H_O\times W_O\times 2}$ is the change classifier and $\sigma(\cdot)$ denotes a softmax function pixel-wisely operated on the channel dimension of the output of the classifier. Our change classifier consists of a convolutional layer of 3×3, in which the number of channels changes from 32 to 2. Finally, by performing a pixel-wise Argmax operation on the channel dimension of M, the final prediction mask $P \in \mathbb{R}^{H_O\times W_O}$ is obtained.

3) Loss Function: During the training phase, we optimize the network parameters by minimize cross-entropy. Formally the loss function is defined as

$$L = \frac{1}{H_p \times W_p}\sum_{m=1,n=1}^{M,N} l(P_{mn}, Y_{mn}) \tag{14}$$

where $l(P_{mn}, y) = -\log(P_{mny})$ is the cross-entropy loss, P_{mn} and Y_{mn} is the label for the pixel at location (h, ω).

IV. Experiments and Analysis

To evaluate the performance of our proposed method, we tested by comparing the results of our model with other CD models on datasets LEVIR-CD and DSIFN-CD. We also designed an ablation experiment to evaluate the effect of our proposed DFMA module, PSM, and transformer-based ST module on us. Finally, we conducted a comprehensive analysis of the experimental results.

A. Datasets

1) LEVIR-CD: Chen and Shi [22] is a new dataset with large-scale RS for building CD. It consists of 637 very high-resolution (VHR, 0.5 m/pixel) Google Earth image patch pairs with dimensions of 1024×1024 pixels. The images include bitemporal images ranging from 5 to 14 years, with significant changes in the complex, which includes villa residences, high-rise apartments, small garages, and large warehouses. In this dataset, the focus will be on building-related changes and building declines. These bidemporal images are annotated using binary labels (1 for variation and 0 for unchanged). The dataset sizes for our train/validation/test are 445/64/128, respectively. During the training process, to improve the processing efficiency, we used random cropping, cropping the picture to 256×256 sizes and small pieces without

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overlapping. During testing, we kept the size of the original dataset and tested the original images of 1024 × 1024 directly. For testing, we used the original dataset and compressed the original 1024 × 1024 images into 256 × 256 directly.

2) DSIFN-CD: Zhang et al. [27] is a dataset collected manually from Google Maps. It consists of six large high-resolution images covering six cities in China, Beijing, Chengdu, Shenzhen, Chongqing, Wuhan, and Xi'an. Here, five large image pairs (i.e., Beijing, Chengdu, Shenzhen, Chongqing, and Wuhan) are cropped into 394 subimage pairs with a size of 512 × 512. And after using data enhancements such as flipping and rotating, a collection of 3988 bitemporal images was obtained. These include marking changes to roads, buildings, farmland and other land cover objects. We trimmed the dataset to 256 × 256 blocks for training, but still followed the original image size for testing. The dataset sizes for our train/validation/test are 3600/300/88, respectively.

3) WHU-CD: Ji et al. [77] is a public CD dataset. It contains a pair of high-resolution (0.2 m) aerial images of size 32507 × 15354. It covers areas that had earthquakes and were rebuilt over several years, mainly building changes. Since the dataset was not processed, we cropped it to 512 × 512 samples for train/validation/test, and further screened too many images that did not produce differences. The dataset sizes for our train/validation/test are 1482/120/138, respectively.

4) CLCD: Liu et al. [78] is a CD dataset consisting of 600 pairs of cropland change sample images. The bitemporal images in it were collected by Gaofen 2 in Guangdong Province, China with 2017 and 2019, respectively, with spatial resolutions from 0.5 to 2 m. Each sample set consists of 512 × 512 images and corresponding binary labels of farmland changes. The training/validation/test size of the dataset was set to 320/120/120, respectively.

B. Metrics an Implementation Deatils

1) Implementation Details: Our model is implemented on pytorch, using a version of torch 1.7, the system using ubuntu, and training using a TITAN RTX graphics card. We performed data enhancements such as rotation, rescaling, cropping, and Gaussian blur on the input training data. We optimized the model using stochastic gradient descent (SGD) with momentum, and set the momentum to 0.99 and the weight attenuation to 0.0005. For the LEVIR-CD, DSIFN-CD, WHU-CD, and CLCD datasets, we set the learning rates to 0.01, 0.0006, 0.01, and 0.01, respectively. After each round of dataset training is completed, the dataset is evaluated using the best model on the validation set.

2) Evaluate Metrics: To evaluate the performance of our method, we mainly use F1-score and interconnection over union (IoU) on variation classification evaluation metrics, in addition to precision (P), recall (R), and overall accuracy (OA). In a more variable task, the higher the precision value, the fewer false detections of the prediction result, and the larger the recall value, the less missed the prediction result. F1-score, IoU, and OA are all composite evaluation indicators. A larger value indicates that the better the prediction. They are represented as follows:

$$\begin{aligned} \mathrm{F1} &= 2\frac{P \cdot R}{P + R} \\ \mathrm{IoU} &= \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN} + \mathrm{FP}} \\ P &= \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \\ R &= \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \\ \mathrm{OA} &= \frac{\mathrm{TP} + \mathrm{pn}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FN} + \mathrm{FP}} \end{aligned} \tag{15}$$

where TP, TN, FP, FN represent the number of true positive, true negative, false positive, and false negative, respectively.

C. Comparison With the Most Advanced Available

In this section, we compare our DMATNet to several of the more advanced methods including three purely convolutional-based methods (fully convolutional (FC)-Siam-Conc [17], FC-Siam-Di [17], FC-early fusion (EF) [17]), three attention-based methods (DSIFN [27], DTCDSCN [26], and SNUNet [31]) and three transformer-based methods (BIT [38], ChangeFormer [79], and CropLand [78]).

1) FC-EF [17] is an image-level approach that is directly based on the network structure of U-Net. Concatenates bitemporal images and processes them through a ConvNet to detect changes.
2) FC-Siam-Di [17] is a characteristic-difference method that extracts multilayered features of bitemporal images from twin networks and uses them for difference detection.
3) FC-Siam-Conc [17] is a feature-concatenation method that extracts multilayered features of bitemporal images from twin networks and uses them for differential detection.
4) DTCDSCN [26] is an attention-based method that uses a dual attention module (DAM) of channel and spatial attention that uses a decoder to link features of different spatial locations and different channels to detect changes.
5) DSIFN [27] using the multiscale feature connection method, the multilevel depth features of the bitemporal image are fused with the image difference features through channel attention and spatial attention. Perform a change map reconstruction.
6) SNUNet [31] is a multilevel feature connection method in which the NestedUNet Siamese network is used for CD and the use of deep supervision to improve the recognition of intermediate features.
7) BIT [38] is a transformer-based method that utilizes a network of converter encoders to enhance the contextual characteristics of the convolution mechanism by working with semantic tokens to obtain change images.
8) ChangeFormer [79] is a transformer-based method that unifies hierarchically structured TE with MLP decoder in a Siamese network architecture.
9) CropLand [78] is a transformer-based method in which a CNN-transformer network with multiscale context

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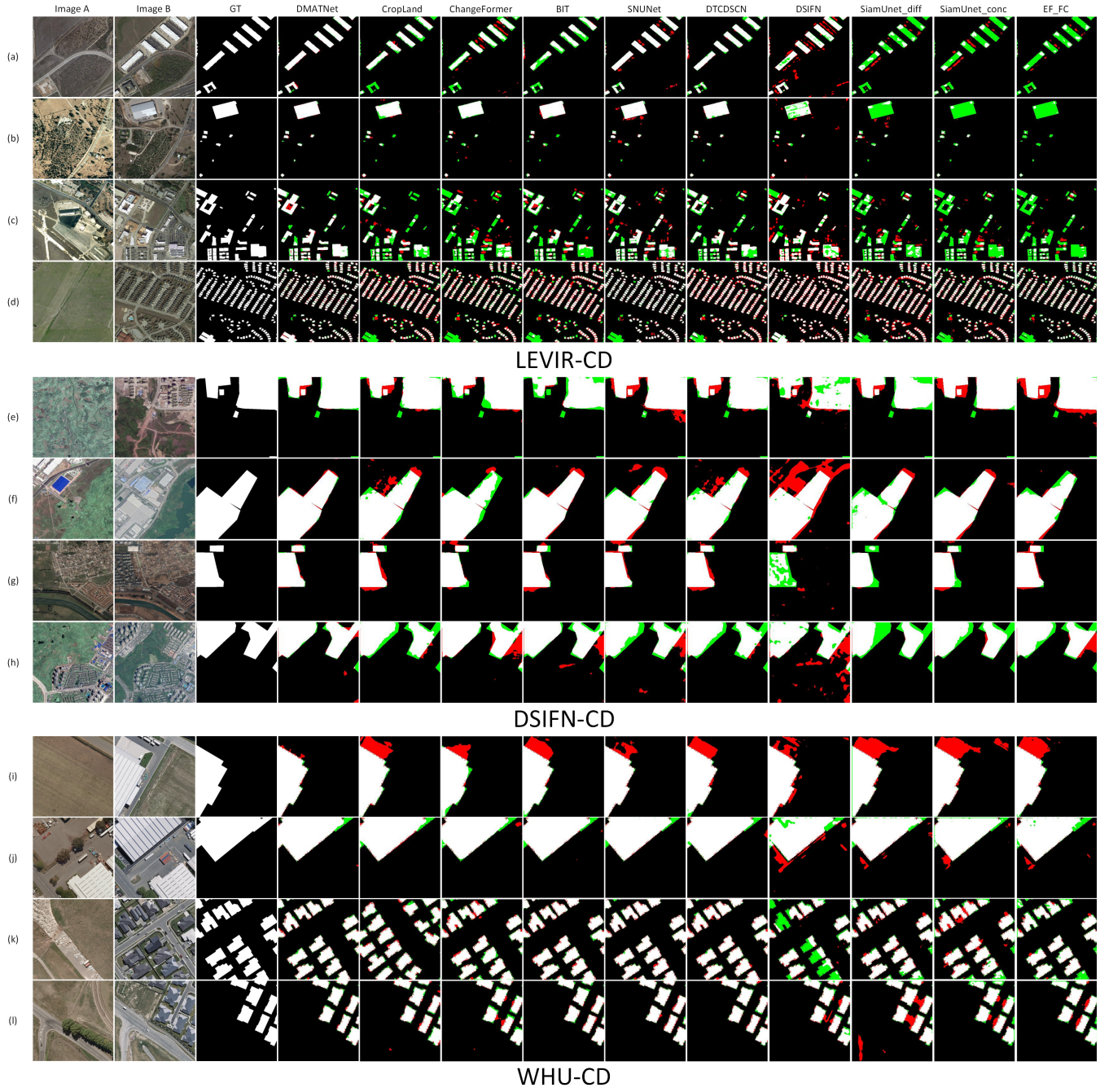


Fig. 6. Visualization results of different methods on the three building CD datasets LEVIR-CD, DSIFN-CD, and WHU-CD. To better distinguish the differences, we use several colors to indicate the different attributes separately. White represents true positives, black represents true negatives, red represents false positives, and green represents false negatives. Where the less green and red, the better the performance represented.

aggregation (MSCANet) is proposed to combine the advantages of CNN and transformer to achieve efficient and effective CD.

The above networks, which we implement using their common code with default parameters, maintain the degree of restoration of the original network as much as possible.

In Table I we give the results of all the CD methods tested on the three building CD datasets. Based on the data in the table, our proposed DMATNet has achieved higher results in indicators such as F1 and IoU, showing better CD performance. Specifically, compared to the existing state-of-the-art transformer-based methods, our F1 and IoU improve by 0.35/1.65, 1.97/2.35, and 0.44/0.67 on the three datasets, respectively, compared to the next highest method. Also, our backbone is based solely on simplified ResNet, with no applications such as feature pyramid networks (FPN) in [22] or Unet used in [26], [27], [31], and [17]. They achieve pixel-level prediction tasks by fusing low-level features, high spatial precision, and high-level semantic features. Therefore,

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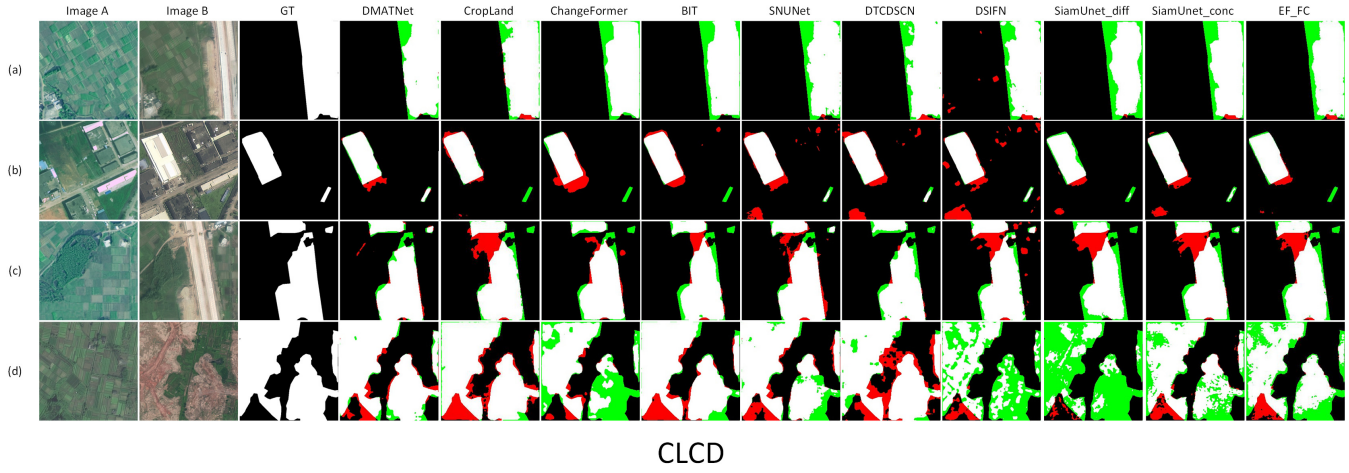


Fig. 7. Visualization results of different methods on the cropland CD dataset CLCD. To better distinguish the differences, we use several colors to indicate the different attributes separately. White represents true positives, black represents true negatives, red represents false positives, and green represents false negatives. Where the less green and red, the better the performance represented.

TABLE I

COMPARE THE RESULTS ON THREE BUILDING CD DATASETS. THE HIGHEST SCORE IS MARKED IN BOLD. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%)

	LEVIR-CD	DSIFN-CD	WHU-CD
	Pre. / Rec. / F1 / IoU / OA	Pre. / Rec. / F1 / IoU / OA	Pre. / Rec. / F1 / IoU / OA
FC-EF [17]	86.91 / 80.17 / 83.4 / 71.53 / 98.39	**72.61** / 52.73 / 61.09 / 43.98 / 88.59	77.69 / 67.10 / 72.01 / 56.26 / 92.07
FC-Siam-Di [17]	89.53 / 83.31 / 86.31 / 75.92 / 98.67	59.67 / 65.71 / 62.54 / 45.50 / 86.63	78.90 / 76.32 / 77.59 / 63.38 / 93.30
FC-Siam-Conc [17]	91.99 / 76.77 / 83.69 / 71.96 / 98.49	66.45 / 54.21 / 59.71 / 42.56 / 87.57	75.41 / 80.04 / 77.66 / 63.47 / 93.00
DTCDSCN [26]	88.53 / 86.83 / 87.67 / 78.05 / 98.77	53.87 / 77.99 / 63.72 / 46.76 / 84.91	87.21 / **82.95** / 85.03 / 73.96 / 95.56
DSIFN [27]	**94.02** / 82.93 / 88.13 / 78.77 / 98.87	67.86 / 53.94 / 60.10 / 42.96 / 87.83	78.00 / 70.81 / 74.23 / 59.03 / 92.53
SNUNet [31]	89.18 / 87.17 / 88.16 / 78.83 / 98.82	60.60 / 72.89 / 66.18 / 49.45 / 87.34	**89.74** / 81.20 / 85.26 / 74.31 / 95.73
BIT [38]	89.24 / 89.37 / 89.31 / 80.68 / **98.92**	68.36 / 70.18 / 69.26 / 52.97 / **89.41**	88.57 / 81.51 / 84.90 / 73.75 / 95.59
ChangeFormer [79]	92.05 / 88.80 / 90.40 / 82.48 / 99.04	57.90 / 72.85 / 64.52 / 47.63 / 87.72	87.25 / 77.03 / 81.82 / 69.24 / 94.80
CropLand [78]	89.79 / 87.57 / 88.67 / 79.64 / 98.86	61.72 / 65.08 / 60.53 / 43.40 / 87.03	83.87 / 75.81 / 79.64 / 66.17 / 94.11
DMATNet	91.56 / **89.98** / **90.75** / **84.13** / 98.25	66.65 / **76.50** / **71.23** / **55.32** / 87.12	89.46 / 82.24 / **85.70** / **74.98** / **95.83**

even our DMATNet model can get superior performance even using a simple backbone network. This may be based on the powerful optimization of features by our PS and DFMA modules, and the idea of using transformers can effectively utilize the space-time information modeling context to enhance feature representation in pixel space.

In Table II, we present the test results on a cropland CD dataset. During the training process, we trained all methods to convergence and tested multiple learning rates to get the best results for each method. On this dataset, our method achieves higher results than the other methods, and DMATNet achieves 6.02/6.46 higher F1 and IoU than the next highest results in the comparison method, which proves that our method can also be applied to CD datasets in other directions with good results.

We visualized the above CD method on three building CD datasets, as shown in Fig. 6. To better visualize the image processing of different CD models, we use four different colors to represent TP (white), TN (black), FP (red), and FN (green). By visualizing the comparison graph, we can observe that our DMATNet model achieves better results than other models.

TABLE II

COMPARISON RESULTS ON THE FARMLAND CD DATASET. THE HIGHEST SCORES ARE MARKED IN BOLD. ALL SCORES ARE DESCRIBED IN PERCENTAGES (%)

	CLCD
	Pre. / Rec. / F1 / IoU / OA
FC-EF [17]	**73.74** / 36.29 / 48.64 / 32.14 / 94.30
FC-Siam-Di [17]	72.97 / 31.60 / 44.10 / 28.29 / 94.04
FC-Siam-Conc [17]	68.21 / 45.22 / 54.48 / 37.35 / 94.35
DTCDSCN [26]	62.98 / 57.53 / 60.13 / 42.99 / 94.32
DSIFN [27]	49.96 / 47.41 / 48.65 / 32.14 / 92.55
SNUNet [31]	65.63 / 56.19 / 60.54 / 43.41 / 94.55
BIT [38]	64.39 / 51.34 / 57.13 / 39.99 / 94.27
ChangeFormer [79]	65.00 / 53.07 / 58.44 / 41.28 / 94.38
CropLand [78]	67.46 / 50.21 / 57.57 / 40.42 / 94.49
DMATNet	72.74 / **61.34** / **66.56** / **49.87** / **95.41**

Due to the presence of PS, our model can gradually narrow the sampling range of feature areas, so our method can better avoid

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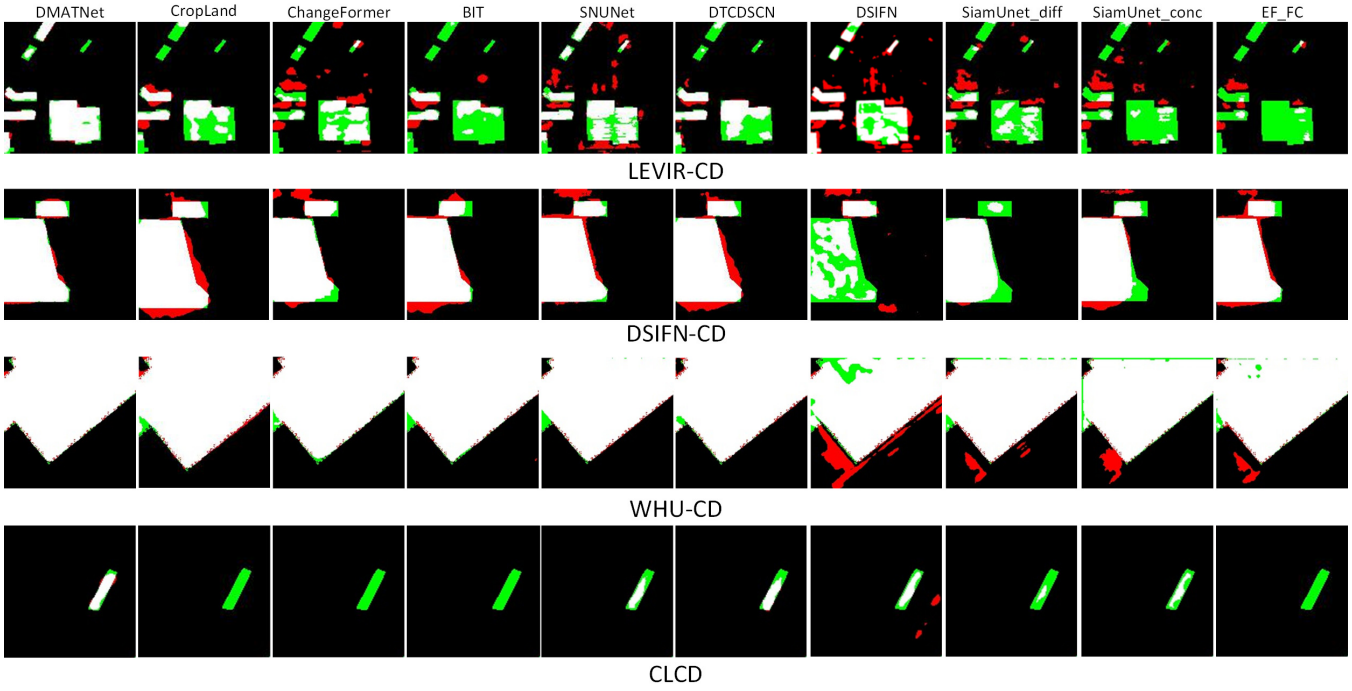


Fig. 8. Visualization of more details on the LEVIR-CD, DSIFN-CD, WHU-CD, and CLCD datasets, where we performed a local zoom on the prediction images to depict the processing details of each method, corresponding to (c), (g), and (j) in Fig 6 and (b) in Fig. 7, respectively.

false positives and false negatives. In the visualization results of the whole comparison experiment, for some buildings that are not easily recognized, such as in (a), (c), and (d), we can not only recognize them effectively but also handle the details well. For images that are easily misidentified, such as in (i), we can effectively target the results to the target area to avoid redundant recognition. Our method also performs well in regionally segmented datasets like DSIFN-CD, and performs well in edge detail processing. It is noteworthy that our method has very few red parts, which is the result we obtained by relying on PSM and DFMA. Moreover, our method has a higher overlap rate compared to other methods.

In Fig. 7, we show the visualization results of all methods on the farmland CD dataset CLCD. From the results, we can see that we also have good results for the detection of target areas such as bare soil, roads, etc. In (b), we can accurately identify the target in the lower right corner, while the other methods either do not identify it or have more redundant parts. In (c), our method has less redundant parts and is very accurate for the top-right target. From the visualization results, we can see that our recognition of the target is still better than the other methods in the comparison experiment.

To show our details more clearly, we have partially enlarged the visual comparison chart, as shown in Fig. 8. These quantitative and qualitative comparisons show that our proposed CD approach is superior to existing CD approaches.

In Table III, we also compare the parameters and time cost of all methods. The results show that our method is in the middle position in terms of the amount of parameters and time cost, and has a good balance between the two.

TABLE III
PARAMETERS AND PROPAGATION TIME OF EACH METHOD TESTED ON THE LEVIR-CD DATASET

	Params	Forward propagation time
FC-EF	1.35M	0.675ms
FC-Siam-Di	1.35M	0.004ms
FC-Siam-Conc	1.55M	0.725ms
DTCDSCN	41.07M	8.624ms
DSIFN	50.71M	6.488ms
SNUNet	12.03M	4.609ms
BIT	12.40M	4.598ms
ChangeFormer	29.75M	2.319ms
CropLand	16.59M	2.111ms
DMATNet	13.27M	4.183ms

D. Ablation Experiments

We conducted partial modular ablation experiments on each of the two datasets. We evaluated all ablation experiments in the same way and performed statistics in Table IV.

1) PS Module: To verify the effect of PS on our network, we performed an ablation experiment by replacing the PSM of the network, that is, directly replacing the PS module with CNN. Since the PS module can extract the image features step by step in an iterative manner, this greatly improves the accuracy of the processing of the edges of the picture. After the lack of PS module, the original token block based on the PS method will not be comprehensive enough for

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TABLE IV

ABLATION EXPERIMENTS WERE PERFORMED ON OUR NETWORK ON TWO CD DATASETS. PS, DFMA, ST, TE, AND TD, RESPECTIVELY. AND ALL METRICS ARE REPORTED. THE DEPTH OF THE PS HERE IS 4. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%)

Model	PS	DFMA	ST	TE	TD	LEVIR-CD Pre. / Rec. / F1 / IoU / OA	DSIFN-CD Pre. / Rec. / F1 / IoU / OA
DMATNet	×	✓	✓	✓	✓	91.16 / 87.87 / 89.43 / 82.24 / 98.05	62.45 / 69.04 / 65.58 / 48.79 / 87.68
DMATNet	✓	×	✓	✓	✓	91.54 / 88.43 / 89.92 / 82.93 / 98.12	67.15 / 57.23 / 63.79 / 44.72 / 86.97
DMATNet	✓	✓	×	✓	✓	86.53 / 79.01 / 82.17 / 73.14 / 96.94	69.29 / 63.58 / 66.32 / 49.61 / 89.02
DMATNet	✓	✓	✓	×	✓	87.69 / 75.60 / 80.35 / 71.02 / 96.82	69.53 / 49.10 / 57.56 / 40.41 / 88.69
DMATNet	✓	✓	✓	✓	×	90.45 / 82.27 / 83.84 / 74.94 / 97.03	69.37 / 64.01 / 66.58 / 49.90 / 87.07
DMATNet	✓	✓	✓	✓	✓	**91.56 / 89.98 / 90.75 / 84.13 / 98.25**	**66.65 / 76.50 / 71.23 / 55.32 / 87.12**

the extraction of spatiotemporal information when the TD is mapped, which will affect the modeling of the context. Therefore, with the absence of the PS module, the data for each evaluation measure has decreased. On both datasets, for our main assessment metrics, F1-score and IoU, the decline was 1.32/2.11, and 3.88/5.18, respectively. For our main evaluation metrics F1 and IoU, the decline was 1.32/2.11 and 5.65/6.53 on the two datasets, respectively. This shows that the PS module has a large role in extracting image features for the token part.

2) Dual-Feature Mixed Attention (DFMA): To verify the effect of the DFMA module on our network, our ablation experimented by deleting the module. In this module, we exchange information about the different features after sampling in two parts. Since PS always relies on the information after the previous sampling, which may lead to some information being unable to be identified, after using DFMA, we can exchange information between the two parts of the feature, which can make up for the disadvantages of the possible inability to identify the part after the gradual sampling. As a result, without this module, our performance has also declined slightly. On both datasets, for our main evaluation metrics, F1 and IoU, the decline was 0.63/0.90 and 4.92/6.21, respectively. For our main evaluation metrics F1 and IoU, the decline is 0.63/0.90 and 7.44/10.60 on the two datasets, respectively. This shows that the DFMA module has a great role in extracting more complete image features.

3) Sampling Tokenizer (ST): We performed an ablation experiment by removing tokenizers. To keep the transformer part unchanged, we consider using dense labeling, that is, extracting the feature sequence from the CNN backbone. The purpose of our ST is to represent features as compactly as possible in tokens, so that the token contains the vast majority of features. As shown in Table IV, our model saw a significant decline in all metrics after the removal of tokenizer. This may be because the dense features sampled by the CNN contain too much redundant information, resulting in a poor distinction between important features and useless features when later mapped back to the original feature. For our main evaluation metrics F1 and IoU, the decline was 8.58/10.99 and 4.91/5.71 on the two datasets, respectively. This shows that the sample marker has an important role to play in our transformer-based framework.

4) Transformer Encoder (TE): We experimented with ablation by removing the encode for tokens. This module serves as a key module in context modeling, and its MSA is a core component in TE. Extracted tokens can be effectively optimized, which is also at the heart of the transformer-based CD framework. As can be seen from Table IV, when our network deleted tees, our data on both datasets dropped significantly. This may be because our network learns context in token-based space, which can be made more compact and have a higher density of information through encoders. For our main evaluation metrics F1 and IoU, the decline was 10.40/10.11 and 13.67/14.91 on the two datasets, respectively. This shows that the TE facilitates efficient extraction between feature relationships.

5) Transformer Decoder (TD): To verify the effectiveness of our TD. We replaced it with a simple fusion module. We calculate by letting the cape of the acquired token expand to, and sum the expanded with feature X, and then pass in the prediction header to generate the final prediction graph. Based on the data in Table IV, we can see that TD is missing and the decline in the indicator is also noticeable, probably because the cross-attention approach provided in TD can enhance the original feature by modeling the relationship between features and using context-rich tags. TD plays a key role in mapping tokens into the original feature. For our main evaluation metrics F1 and IoU, the decline was 6.91/9.19 and 4.65/5.42 on the two datasets, respectively.

E. Parameter Analysis for Key Modules

In this section, we tested PS depths on two datasets, LEVIR-CD and DSIFN-CD to verify that our model has obtained a local optimal solution on this module.

1) Depth of the PS: Our PS helps with token extraction later on, as well as the processing of the transformer. The depth of PS (N_{PS}) is an important parameter to extract more precise features. We tested the change in our network metric score when the PS configuration of $N_{PS} = 1, 2, 4, 6$ was performed. The experimental parameters are carried out in order from smallest to largest, and the test results are compared directly by the main measure f1-score and IoU, as shown in Table V. On the LEVIR-CD dataset, when the number of layers is 1 and 2, the effect obtained is even lower than that of the PS module, which indicates that in the case of insufficient sampling, the PS module is not good for the extraction of

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TABLE V
EFFECT OF PS LAYERS ON THE MODEL IS TESTED. WE REPORTED F1-SCORE AND IOU ON DMATNET AT DIFFERENT DEPTHS. WHERE PS.D REPRESENTS THE DEPTH OF PS. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%)

	LEVIR-CD		DSIFN-CD	
PS.D	F1	IoU	F1	IoU
1	83.81	74.94	64.32	48.93
2	84.80	76.16	**71.31**	**55.46**
4	**90.75**	**84.13**	71.23	55.32
6	89.49	82.32	60.87	45.76

features, and may extract the wrong feature information, thus affecting the mapping of the later tokens. During the increase in the number of layers from 1 to 4, our evaluation index continues to rise, which may be because as the sampling depth increases, the extraction of features becomes more refined, thereby reducing the impact of redundant features. However, the effect decreases as the depth continues to increase from 4 to 6, either because too many iterations may lose some useful feature information, or because the information is too complex during the extraction process, which affects the effective mapping of the token to the original feature. On the DSIFN-CD dataset, when $N_{PS} = 2$, the best results are achieved. This may be due to the slight decrease in image detail in the DSIFN-CD (2 m/pixel) dataset compared to the LEVIR-CD (0.5 m/pixel) dataset, and the mask of the former is mostly a holistic segmentation of the complex rather than each building. Therefore, when the PS depth is too deep, it will cause some of the sampling information to be incorrectly ignored, which will lead to better sampling performance when the sampling depth is shallow. However, since the depths of 2 and 4 have less impact on the DSIFN-CD dataset, far less than the LEVIR-CD dataset, we will set N_{PS} to 4.

V. DISCUSSION

We propose a DFMA that can adequately mix coarse and fine features, which greatly helps the detail processing of RS images. From the tested data, our IOU has higher values compared with other networks, but OA is often low. The impact is clearly felt in lower spatial resolution datasets, such as the DSIFN-CD dataset. The main reason should be that when PS concentrates the picture-specific attention to a certain region, if the other feature extraction method extracts too large a range so that it differs significantly from the PSM results, it is likely to fail to play a better complementary role to each other. In contrast, with higher spatial resolution, the two features do not differ much, which will play a better role in facilitating the final fused features. Besides, DSIFN-CD is divided based on a piece of house area as the target, while LEVIR-CD and WHU-CD are targeting buildings and detecting different targets, so the details of the images are not very obvious on DSIFN-CD. However, according to the experimental data, it is not difficult to find that our effect is more obvious for higher resolution datasets, and in the future when the satellite equipment is more advanced, our method will also have good effect on high-resolution datasets. Also, we will try new feature extraction methods in DFMA to improve the method with better results.

VI. CONCLUSION

In this article, we propose DFMA-based transformer CD model. In which, we use an iterative PS to further optimize the features and fully interact the feature information in DFMA so that the final information with more efficient is extracted. The blended feature information is then optimized using the transformer-based CD framework for feature information, making the contextual information more comprehensive. We have conducted a large number of experiments to prove the effectiveness of this method and have obtained significant accuracy improvements. Our experimental evidence suggests that DMATNet is more efficient than the most advanced methods based on pure convolution and pure attention mechanisms, and has better results in detail than existing transformer methods, especially on datasets with high spatial resolution. In the future, with the availability of high-precision datasets, our method will also show certain advantages.

REFERENCES

[1] A. P. Tewkesbury, A. J. Comber, N. J. Tate, A. Lamb, and P. F. Fisher, "A critical synthesis of remotely sensed optical image change detection techniques," *Remote Sens. Environ.*, vol. 160, pp. 1–14, Apr. 2015.
[2] M. Hussain, D. Chen, A. Cheng, H. Wei, and D. Stanley, "Change detection from remotely sensed images: From pixel-based to object-based approaches," *ISPRS J. Photogramm. Remote Sens.*, vol. 80, pp. 91–106, Jun. 2013.
[3] L. Bruzzone and D. F. Prieto, "Automatic analysis of the difference image for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1171–1182, May 2000.
[4] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.
[5] J. S. Deng, K. Wang, Y. H. Deng, and G. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, 2008.
[6] C. Wu, B. Du, X. Cui, and L. Zhang, "A post-classification change detection method based on iterative slow feature analysis and Bayesian soft fusion," *Remote Sens. Environ.*, vol. 199, pp. 241–255, Sep. 2017.
[7] C. Huang et al., "Use of a dark object concept and support vector machines to automate forest cover change analysis," *Remote Sens. Environ.*, vol. 112, no. 3, pp. 970–985, 2008.
[8] K. Tan, X. Jin, A. Plaza, X. Wang, L. Xiao, and P. Du, "Automatic change detection in high-resolution remote sensing images by using a multiple classifier system and spectral–spatial features," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 8, pp. 3439–3451, Aug. 2016.
[9] M. Volpi, D. Tuia, F. Bovolo, M. Kanevski, and L. Bruzzone, "Supervised change detection in VHR images using contextual information and support vector machines," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 20, pp. 77–85, Feb. 2013.
[10] G. Chen, G. J. Hay, L. M. T. Carvalho, and M. A. Wulder, "Object-based change detection," *Int. J. Remote Sens.*, vol. 33, no. 14, pp. 4434–4457, 2012.
[11] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, May 2015.

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[12] Y. Sun, H. Song, A. J. Jara, and R. Bie, "Internet of Things and big data analytics for smart and connected communities," *IEEE Access*, vol. 4, pp. 766–773, 2016.

[13] J. Yang, T. Liu, B. Jiang, H. Song, and W. Lu, "3D panoramic virtual reality video quality assessment based on 3D convolutional neural networks," *IEEE Access*, vol. 6, pp. 38669–38682, 2018.

[14] J. Liu, M. Gong, K. Qin, and P. Zhang, "A deep convolutional coupling network for change detection based on heterogeneous optical and radar images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 3, pp. 545–559, Mar. 2016.

[15] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on deep Siamese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Aug. 2017.

[16] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Auto. Robots*, vol. 42, no. 7, pp. 1301–1322, 2018.

[17] R. C. Daudt, B. L. Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[18] M. Papadomanolaki, S. Verma, M. Vakalopoulou, S. Gupta, and K. Karantzalos, "Detecting urban changes with recurrent neural networks from multitemporal Sentinel-2 data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 214–217.

[19] L. Xu, W. Jing, H. Song, and G. Chen, "High-resolution remote sensing image change detection combined with pixel-level and object-level," *IEEE Access*, vol. 7, pp. 78909–78918, 2019.

[20] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[21] M. Zhang, G. Xu, K. Chen, M. Yan, and X. Sun, "Triplet-based semantic relation learning for aerial remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 266–270, Feb. 2019.

[22] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[23] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.

[24] M. Zhang and W. Shi, "A feature difference convolutional neural network-based change detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7232–7246, Oct. 2020.

[25] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," 2014, *arXiv:1409.0473*.

[26] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[27] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[28] X. Peng, R. Zhong, Z. Li, and Q. Li, "Optical remote sensing image change detection based on attention mechanism and image difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7296–7307, Sep. 2021.

[29] H. Jiang, X. Hu, K. Li, J. Zhang, J. Gong, and M. Zhang, "PGA-SiamNet: Pyramid feature-based attention-guided Siamese network for remote sensing orthoimagery building change detection," *Remote Sens.*, vol. 12, no. 3, p. 484, Feb. 2020.

[30] F. I. Diakogiannis, F. Waldner, and P. Caccetta, "Looking for change? Roll the dice and demand attention," *Remote Sens.*, vol. 13, no. 18, p. 3707, 2021.

[31] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[32] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 6000–6010.

[33] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[34] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890.

[35] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[36] X. Chu et al., "Twins: Revisiting the design of spatial attention in vision transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 9355–9366.

[37] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.

[38] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[39] Y. Feng, H. Xu, J. Jiang, H. Liu, and J. Zheng, "ICIF-Net: Intra-scale cross-interaction and inter-scale feature fusion network for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[40] H. Zhang, M. Gong, P. Zhang, L. Su, and J. Shi, "Feature-level change detection using deep representation and feature change analysis for multispectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 11, pp. 1666–1670, Nov. 2016.

[41] S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised deep change vector analysis for multiple-change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3677–3693, Jun. 2019.

[42] B. Hou, Y. Wang, and Q. Liu, "Change detection based on deep features and low rank," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2418–2422, Dec. 2017.

[43] D. Peng and H. Guan, "Unsupervised change detection method based on saliency analysis and convolutional neural network," *J. Appl. Remote Sens.*, vol. 13, no. 2, p. 1, May 2019.

[44] Y. Lei, X. Liu, J. Shi, C. Lei, and J. Wang, "Multiscale superpixel segmentation with deep features for change detection," *IEEE Access*, vol. 7, pp. 36600–36616, 2019.

[45] A. M. El Amin, Q. Liu, and Y. Wang, "Zoom out CNNs features for optical remote sensing change detection," in *Proc. 2nd Int. Conf. Image, Vis. Comput. (ICIVC)*, Jun. 2017, pp. 812–817.

[46] P. de Bem, O. de Carvalho Junior, R. F. Guimarães, and R. T. Gomes, "Change detection of deforestation in the Brazilian Amazon using landsat data and convolutional neural networks," *Remote Sens.*, vol. 12, no. 6, p. 901, Mar. 2020.

[47] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. XLII-2, pp. 565–571, May 2018.

[48] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, 2019.

[49] T. Bao, C. Fu, T. Fang, and H. Huo, "PPCNET: A combined patch-level and pixel-level end-to-end deep network for high-resolution remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 10, pp. 1797–1801, Oct. 2020.

[50] B. Hou, Q. Liu, H. Wang, and Y. Wang, "From W-Net to CDGAN: Bitemporal change detection via deep learning techniques," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1790–1802, Mar. 2020.

[51] B. Fang, L. Pan, and R. Kou, "Dual learning-based Siamese framework for change detection using bi-temporal VHR optical remote sensing images," *Remote Sens.*, vol. 11, no. 11, p. 1292, 2019.

[52] M. Liu, Q. Shi, A. Marinoni, D. He, X. Liu, and L. Zhang, "Super-resolution-based change detection network with stacked attention module for images with different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[53] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[54] Z.-G. Liu, Z.-W. Zhang, Q. Pan, and L.-B. Ning, "Unsupervised change detection from heterogeneous data based on image translation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[55] H. Chen, W. Li, and Z. Shi, "Adversarial instance augmentation for building change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[56] F. Luo, Z. Zou, J. Liu, and Z. Lin, "Dimensionality reduction and classification of hyperspectral image via multistructure unified discriminative embedding," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[57] Y. Zhang, K. Wu, B. Du, and X. Hu, "Multitask learning-based reliability analysis for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2135–2147, Jul. 2019.

[58] S. Li and L. Huo, "Remote sensing image change detection based on fully convolutional network with pyramid attention," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 4352–4355.

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[59] X. Li, Z. Du, Y. Huang, and Z. Tan, "A deep translation (GAN) based change detection network for optical and SAR remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 14–34, Sep. 2021.
[60] M. Chen et al., "Generative pretraining from pixels," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1691–1703.
[61] P. Esser, R. Rombach, and B. Ommer, "Taming transformers for high-resolution image synthesis," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12873–12883.
[62] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 10347–10357.
[63] B. Wu et al., "Visual transformers: Token-based image representation and processing for computer vision," 2020, *arXiv:2006.03677*.
[64] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.
[65] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.
[66] W. Liu, S. Chen, L. Guo, X. Zhu, and J. Liu, "CPTR: Full transformer network for image captioning," 2021, *arXiv:2101.10804*.
[67] F. Yang, H. Yang, J. Fu, H. Lu, and B. Guo, "Learning texture transformer network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5791–5800.
[68] H. Chen et al., "Pre-trained image processing transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12299–12310.
[69] Y. Yuan and L. Lin, "Self-supervised pretraining of transformers for satellite image time series classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 474–487, 2021.
[70] Z. Li, G. Chen, and T. Zhang, "A CNN-transformer hybrid approach for crop classification using multitemporal multisensor images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 847–858, 2020.
[71] X. Shen, B. Liu, Y. Zhou, and J. Zhao, "Remote sensing image caption generation via transformer and reinforcement learning," *Multimedia Tools Appl.*, vol. 79, nos. 35–36, pp. 26661–26682, Sep. 2020.
[72] Q. Wang, W. Huang, X. Zhang, and X. Li, "Word–sentence framework for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10532–10543, Dec. 2021.
[73] Y. Bazi, L. Bashmal, M. M. A. Rahhal, R. A. Dayil, and N. A. Ajlan, "Vision transformers for remote sensing image classification," *Remote Sens.*, vol. 13, no. 3, p. 516, 2021.
[74] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Sep. 2019.
[75] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.
[76] X. Yue et al., "Vision transformer with progressive sampling," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 387–396.
[77] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.
[78] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4306, 2022.
[79] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," 2022, *arXiv:2201.01293*.



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